

of any of the SUBEVENTS of EVALUATE-FLAG-USER-MOVE. If the preconditions for detecting any of these error types are met, the agent issues an associated message.

EVALUATE-FLAG-USER-MOVE is an intermediate node in the ontology whose description is useful if someone asks an agent a metalevel question like *What do tutors do?* or *What kinds of mistakes do tutors look out for?* However, in order for the agent to perform as a tutor or advisor in an application, it needs domain-specific knowledge, which is recorded in the appropriate descendant of TUTORING-ADVISING-SCRIPT, for example:

TUTORING-ADVISING-SCRIPT
TUTORING-ADVISING-**CLINICAL-MED**
TUTORING-ADVISING-**FURNITURE-ASSEMBLY**
TUTORING-ADVISING-**DRIVING-A-VEHICLE**

Taking the example of clinical medicine, when a tutoring or advising session for clinical medicine begins, the agent places an instance of TUTORING-ADVISING-**CLINICAL-MED** on its agenda. One of its subevents is EVALUATE-FLAG-**CLINICAL-MED**-MOVE, which holds the knowledge about how to respond specifically to actions in the domain of clinical medicine.

EVALUATE-FLAG-**CLINICAL-MED**-MOVE
DEFINITION This script evaluates user moves in the domain of clinical medicine and detects different classes of errors, including those that might result from cognitive biases.

AGENT LEIA
BENEFICIARY HUMAN
SUBEVENTS
TRY: DETECT-PLAIN-ERROR-**MED**
TRY: DETECT-JUMPING-TO-CONCLUSIONS-**MED**
TRY: DETECT-FRAMING-SWAY-**MED**
TRY: DETECT-SMALL-SAMPLE-BIAS-**MED**
TRY: DETECT-BASE-RATE-NEGLECT-**MED**
TRY: DETECT-ILLUSION-OF-VALIDITY-**MED**
TRY: DETECT-EXPOSURE-EFFECT-**MED**

For purposes of illustration, we will describe two of these subevents: the one that detects jumping to conclusions, which we introduced in passing earlier; and the one that detects presenting information using a framing sway, which involves phrasing it in a way that could subtly influence the hearer's response to it.

Agents detect jumping to conclusions by comparing user moves against the preconditions of good practice recorded in the ontology. Good practice is encapsulated in guidelines, agreed upon by clinicians, that inform clinical decision-making. For example, there are guidelines for determining when there is enough evidence to hypothesize or diagnose a disease, and when it is justified to recommend tests and interventions. Table 8.5 shows some of the preconditions of good practice recorded in the ontological description of the disease achalasia. For readability, the fillers are described using English strings rather than meaning representations written in the ontological metalanguage.

Using this information, agents functioning as tutors and advisors can detect and flag if users are making moves that are not yet clinically justified, and they can answer questions like *Can a diagnosis be made yet?*

A knowledge-engineering aside. Tables like 8.5 served as the common ground between knowledge engineers and physician educators during development of the Maryland Virtual Patient system, and they are useful in presenting cognitive models to students, educators, and other stakeholders.

Although we already presented the algorithm for detecting jumping to conclusions about diagnoses, we repeat it here for easy comparison with table 8.5, in order to emphasize how agents leverage knowledge while evaluating decision functions.

Table 8.5
Some of the preconditions of good practice related to the disease achalasia.

Property	English gloss of filler
SUFFICIENT-FOUNDATIONS-TO-SUSPECT	Dysphagia to solids and liquids or regurgitation
SUFFICIENT-FOUNDATIONS-TO-DIAGNOSE	All four of the following conditions: 1. either bird’s beak or lower esophageal sphincter pressure > 45 2. aperistalsis 3. either dysphagia or regurgitation 4. negative EGD for cancer
SUFFICIENT-FOUNDATIONS-TO-TREAT	Definitive diagnosis of achalasia
PREFERRED-ACTION-WHEN-DIAGNOSED	Pneumatic dilation or Heller myotomy
REASONABLE-ACTION-WHEN-DIAGNOSED	Administer BoTox

DETECT-JUMPING-TO-CONCLUSIONS-DIAGNOSIS

DEFINITION This procedure detects if a doctor has sufficient evidence to make a diagnosis.

AGENT LEIA

SUBEVENTS

TRY: sufficient-grounds-to-diagnose-ok

EXPL "Diagnosing [disease] is clinically valid. The relevant feature values are [contributing-feature-values]."

TRY: no-sufficient-grounds-to-diagnose

EXPL "Diagnosing [disease] is not clinically valid. [contributing-feature-values] contribute to making the diagnosis but [absent-feature-values] must also be known."

If a user diagnoses a disease, the agent evaluates the move against its knowledge of good clinical practices. If the move is valid, then the agent simply records it, including why it was justified, in case the agent is asked about it later on. The justification is a trace of the preconditions that were fulfilled in order to satisfy the conditions of the function `sufficient-grounds-to-diagnose-ok`.

If, by contrast, a user move is incorrect, then the agent instantiates the concept `FLAG-CLINICAL-MOVE`, which is the output of the function `no-sufficient-grounds-to-diagnose`, and passes to that function the information about which move was erroneous and why.

Consider the following example that shows how a LEIA playing the role of a tutor responds when a clinician in training posits a diagnosis without sufficient evidence.

- The student says to the virtual patient *You have achalasia*.
- The LEIA's language understanding system understands this to mean "DIAGNOSE-DISEASE-1 (THEME ACHALASIA-1) (ACHALASIA-1 (EXPERIENCER PATIENT-1))."
- Since `TUTORING-ADVISING-CLINICAL-MED` is on the agent's agenda, given the agent's role as a tutor, the agent tests everything it perceives against all of that plan's `SUBEVENTS`, one of which is `EVALUATE-FLAG-CLINICAL-MED-MOVE`.
- We assume for this example that the diagnosis matches the second condition, `no-sufficient-grounds-to-diagnose`, because although the first three conditions of `SUFFICIENT-GROUNDS-TO-DIAGNOSE` in table 8.5 are fulfilled, the fourth one is not: negative EGD for cancer.
- The LEIA records relevant metadata associated with this decision:
 - `CONTRIBUTING-FEATURE-VALUES` (in plain English): a bird's beak, aperistalsis, and dysphagia.
 - `ABSENT-FEATURE-VALUES` (in plain English): negative EGD for cancer.

- The LEIA instantiates FLAG-CLINICAL-MOVE, and its language generation system outputs: “Diagnosing achalasia is not clinically valid. A bird’s beak, aperistalsis, and dysphagia contribute to making the diagnosis but a negative EGD for cancer must also be known.”⁷

Dedicated algorithms relying on ontological knowledge similar to that in table 8.5 are available to the agent for evaluating whether the preconditions of good practice have been fulfilled for recommending tests and procedures as well.

Which kind of warning the agent generates depends on the type of user, student vs. professional, as well as user preferences about how much information to provide when issuing warnings. For example, in the Maryland Virtual Patient application, students could choose to see no tutoring messages, minimalistic messages (e.g., there is insufficient evidence to diagnose a disease), messages with context-specific information (e.g., what else must be known in order to diagnose the disease in the given patient), or messages with extensive additional information (e.g., all of the different ways of fulfilling the preconditions for diagnosing the given disease). The information-rich option is illustrated by figure 8.4 in section 8.7, which shows all clinically justified grounds for ordering the test called EGD as well as which preconditions were already satisfied at the given point in the given simulation run.

The second example of a flaggable user move that we will consider involves linguistic priming. In clinical scenarios, the way a doctor describes interventions, presents options, and asks questions can impact patients’ impressions and their subsequent decision-making. For example,

- If the doctor asks “I imagine your throat hurts, right?,” the patient will have a tendency to seek corroborating evidence, even if he or she had not previously noticed any throat pain. This is the *confirmation bias*.
- If the doctor asks “Your pain is very bad, isn’t it?,” the patient is likely to overestimate the perceived pain, having been primed with a high pain level. This is the *priming effect*.
- If the doctor says, “There is a 20% chance that this procedure will fail,” the patient is likely to interpret the procedure more negatively than if the doctor had said, “There’s an 80% chance that this will succeed.” This is the *framing sway*.

Agents can help doctors to be aware of, and learn to avoid, such formulations by flagging potentially bias-inducing utterances. The detection